IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 11 August 2025, accepted 2 September 2025,
date of publication 9 September 2025, date of current version 16 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3607746

RESEARCH ARTICLE

ARQ_HMAC: An Automatic Repeat reQuest Strategy for Hybrid MAC Protocol in Underwater Acoustic Sensor Networks

VAN-PHUONG DAO AND THU-HA TRAN
Faculty of Electrical and Electronics Engineering, Ho Chi Minh City University of Technology and Education, Ho Chi Minh City 700000, Vietnam
Corresponding author: Van-Phuong Dao (phuongdv@hcmute.edu.vn)

This work belongs to the project in 2025 funded by Ho Chi Minh City University of Technology and Education, Vietnam.

ABSTRACT In underwater sensor networks, the use of acoustic waves with large propagation delays and narrow channel bandwidth poses significant challenges for Medium Access Control (MAC) protocols in reducing the number of retransmissions caused by packet collisions at the receiver node, thereby improving channel throughput. When packet collisions occur, the transmitting node enters a waiting state and will retransmit the packet at an appropriate time. However, if a node retransmits the packet at inappropriate times, it can lead to further collisions in subsequent transmission attempts. To reduce the number of retransmissions caused by collisions and improve channel throughput, in this study, we propose a Hybrid MAC protocol that combines asynchronous contention-free MAC with Carrier Sense Multiple Access (CSMA) and synchronous Time Division Multiple Access (TDMA) MAC protocols, along with an ARQ strategy applied to underwater communication channels, called the ARQ_HMAC (Automatic Repeat reQuest Strategy for Hybrid MAC) strategy. In the ARQ_HMAC protocol, when a node has a packet to transmit, it senses the channel state and operates according to the Contention-Free MAC protocol. If a packet collision occurs, the node switches to a waiting state and retransmits the packet using the TDMA protocol. The waiting time and the TDMA schedule are calculated based on the ARQ_HMAC strategy to avoid collisions in subsequent retransmission attempts. The simulation results will demonstrate that the ARQ_HMAC protocol helps reduce the number of packet retransmissions caused by collisions, thereby increasing and stabilizing the channel throughput as the arrival rate of new packets increases.

INDEX TERMS Underwater communication channel, contention-free MAC protocol, TDMA MAC protocol, hybrid MAC protocol, automatic repeat request strategy.

I. INTRODUCTION

Underwater wireless communication networks consist of a variable number of sensor nodes or vehicles that are deployed to perform collaborative monitoring tasks or collect scientific data over a given area. The sensor nodes or vehicles gather underwater information and transmit this information, either through a sequence of relays or directly, to a master node, which then forwards the information to shipboard data processing centers or other networks as show in Fig.1. To achieve this objective, sensor nodes or vehicles self-organize in an autonomous underwater network which can adapt to the characteristics of the underwater environment.

In terrestrial wireless communication networks, radio waves are used; however, these are not suitable for underwater wireless communication networks. Instead, acoustic waves are used for communication in underwater wireless channels. The unique characteristics of this environment must be considered, such as slower propagation speed and narrower bandwidth. Specifically, the propagation velocity of acoustic waves underwater is approximately 1500 m/s, with a bandwidth of only a few kilohertz and multipath propagation effects significantly lower than those of radio waves used in terrestrial communication [1]. The slow propagation

The associate editor coordinating the review of this manuscript and approving it for publication was Ananya Sen Gupta.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

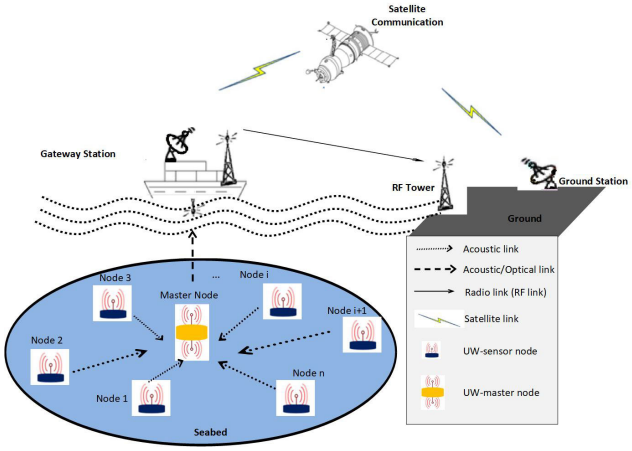


FIGURE 1. Network configuration.

speed of acoustic waves in underwater acoustic channels leads to increased packet collisions, thereby reducing the overall performance of the communication channel [2]. Many channel access protocols with ARQ strategies have been proposed to address the limitations of acoustic channels, including improved versions of existing protocols, with the aim of minimizing packet collisions during transmission and improving channel efficiency [3], [4], [5], [6]. MAC protocols designed for underwater communication channels are typically categorized into three main groups: contention-free, contention-based, and hybrid MAC protocols [7].

Contention-free MAC protocols include methods such as Frequency Division Multiple Access (FDMA). However, due to the limited bandwidth of underwater acoustic channels and their susceptibility to fading and multipath effects, the straightforward application of FDMA is not well-suited for underwater acoustic communication [8].

For the Time Division Multiple Access (TDMA) protocol, a careful design of time slots is required to ensure separation between different transmissions and to reduce the probability of packet collisions during data transmission and reception periods. This is especially critical due to the significant propagation delays and delay variances inherent in acoustic channels. These requirements can lead to lower channel utilization. Furthermore, the variable delays make precise synchronization difficult, which is essential for effective TDMA operation [9].

Code Division Multiple Access (CDMA) is a medium access protocol that does not require synchronization. However, due to the limitations of CDMA - particularly the near-far problem, which poses a significant design challenge for underwater MAC protocols, there have been few developments in contention-free CDMA-based MAC protocols. To address the near-far problem, transmission power control strategy are employed to reduce the transmission power of each node [8].

In the categorization of contention-based MAC protocols for underwater communication channels, there are typically two main approaches to random channel access: Carrier Sense Multiple Access (CSMA) and the original ALOHA protocol, along with many variations, as noted in [10] and [11]. In practical underwater communication, due to the low propagation speed of acoustic waves, the hidden and exposed node problems present significant challenges for random access methods. With ALOHA-based schemes, transmitting nodes do not consider the channel state before sending their data, which can lead to a high probability of packet collisions.

Regarding handshaking-based protocols [12], an approach involving the exchange of Request to Send (RTS) and Clear to Send (CTS) control packets is used to reserve the channel prior to data transmission. While this method can help reduce collisions, it may also result in slower data transmission speeds due to the additional time required for RTS/CTS exchanges.

CSMA performs efficiently in fully connected networks where propagation delays are small relative to the packet duration. However, as the delay increases, its efficiency declines rapidly. Furthermore, in AD-HOC networks, the issues of hidden and exposed terminals occur due to the absence of connectivity between certain nodes. On the other hand, CSMA schemes may result in excessively long vulnerable periods, and the cost of carrier sensing in underwater environments is relatively high.

Channel access protocols play a crucial role in reducing packet collisions and improving overall channel performance. However, due to the complex characteristics of underwater transmission environments, handshake-based protocols can only minimize packet collisions to a certain extent-collisions may still occur [9].

Contention-free MAC protocols do not employ any collision avoidance mechanisms prior to transmission, allowing them to make efficient use of channel resources for packet delivery; however, the packet collision rate becomes high as the offered load increase [7].

Contention-based MAC protocols employ a handshake or synchronization process before transmission, which significantly reduces the likelihood of packet collisions. However, part of the channel resources is consumed by the handshake process, thereby limiting channel utilization and consequently reducing the overall throughput of the channel [17].

When packet collisions happen, nodes that are preparing to transmit may sense the channel as busy, prompting them to enter a sleep state and wait for a specific duration before attempting retransmission. If the colliding packets are retransmitted at the same intervals, repeated collisions can occur, potentially leading to network paralysis. Therefore, an effective strategy for managing nodes' sleep and wake cycles-particularly upon detecting a busy channel or failed transmission is essential for alleviating network congestion caused by persistent packet collisions.

The remainder of this paper is structured as follows. Section II reviews the related work. Section III provides a detailed description of the proposed ARQ_HMAC

strategy. Section IV presents the analysis of network throughput. Section V outlines the simulation setup, including parameters, execution procedures, and the presentation and analysis of the results. Finally, Section VI summarizes the main findings and concludes the paper.

II. RELATED WORKS

Hybrid MAC protocols combine various medium access control methods in an adaptive and traffic-aware manner to improve communication performance in underwater acoustic networks.

A hybrid MAC protocol include hierarchical multi-channel MAC protocols was proposed by Diamant et al., [13]. This study presented a MAC protocol for underwater acoustic communication networks that applies spatial reuse to improve network throughput. In the proposed protocol selected additional nodes can transmit simultaneously to the active TDMA node to improve the efficiency of the MAC protocol and optimizes the set of active nodes and overcomes problems of underwater acoustic communication networks such as the near-far problem and flickering.

A Hybrid NOMA-Based MAC (HN-MAC) Protocol for underwater acoustic networks have been proposed in the study by Guo et al. [14]. HN-MAC solves the problem of concurrent communication for underwater acoustic networks which combines power-domain and code-domain NOMA to classify users and allocate communication resources. For the user classification, an Adaptive Clustering Algorithm (ACA) was proposed, which dynamically determines the clusters' number and classifies users based on channel gain and channel correlation under multipath conditions. HN-MAC can decrease interference among multiple users.

Namgung et al. [15] conducted a study on Preamble-MAC (P-MAC) protocol. This protocol is adaptive and dynamic MAC protocol based on virtual distance level, which is the estimated, accumulated information of channel status and variation obtained through periodically monitoring underwater environment information such as long, unknown propagation delays and low bandwidth for underwater acoustic sensor networks.

In the study of utilizing CDMA as the underlying multiple access technique, Watfa et al. [16] proposed UW-MAC, a CDMA-based power-controlled medium access protocol that utilizes both transmitter-based and receiver-based CDMA inside a formed cluster, and utilizes a TDMA schedule to create the cluster heads communicate with the base station. This MAC protocol targets to increase the overall network lifetime.

Roy et al. [17] proposed a synchronous duty-cycled reservation-based MAC protocol called Ordered Contention MAC (OCMAC) protocol. The basic idea of this protocol is to suggest a schedule data transmission by transmitting nodes through the scheduling of Ready To Send (RTS) frames. The OCMAC protocol can eliminates the possible collision during data transmission and improves communication efficiency.

A Multichannel Ordered Contention MAC (MOC-MAC) protocol enhances underwater wireless sensor networks (UWSNs) by utilizing multiple channels for data and control packets was introduced. In this study, Roy et al. [18] want to reduce collisions and improve throughput in bursty traffic scenarios. This protocol builds upon the Ordered Contention MAC (OCMAC) by incorporating multichannel communication, enabling nodes to transmit data on different channels simultaneously, thus mitigating contention and improving network performance.

Roy et al. [19] proposed an asynchronous receiver-initiated preamble-based MAC protocol called Receiver Preambling with Channel Polling MAC (RPCP-MAC) protocol for shallow underwater monitoring applications with high data rates. RPCP-MAC protocol resolves data packet collision and support reliability in an energy-efficient way without using any transmission schedule. The protocol is primary based on the mechanisms: the receiver preambling mechanism is adopted to reduce idle listening; then, channel polling mechanism is used to determine missing data frame during its sleeping period and to minimize the active time of node and reduces energy wastage. Finally, a back-off mechanism is applied to resolve collision when preambles are received simultaneously.

A hybrid MAC protocol with centralized scheduling and distributed scheduling strategies is proposed in this study by Zhao et al. [20]. In this MAC protocol, the scheduling strategies which is the centralized scheduling for sensor nodes with periodic data and the distributed scheduling strategy for sensor nodes with intermittent data. And the scheduling strategy used by a sensor node is determined by its data rate. The proportion of resource allocated for the centralized scheduling and the distributed scheduling is determined by the sink node according to the network load.

Hybrid MAC protocols leverage the strengths of different MAC techniques while addressing their limitations to offer more efficient solutions. However, these approaches may face design challenges related to increased communication overhead, primarily due to the complexity of the integrated algorithms.

The aforementioned MAC protocols focus on identifying and solving the issue of packet collisions. In the event of a collision, the nodes either switch to a sleep mode and set a random wake-up time to retransmit the packet, which results in a high probability of subsequent collisions after the first one. Several studies on ARQ strategies have focused on ARQ computation, such as: A study of ARQ was presented by Stojanovic et al., [21] includes the Stop-and-Wait (S&W) approach, in which a sending node must wait for an acknowledgment of the previous packet's successful transmission before sending the next one.

Several improvements to the Stop-and-Wait protocol have been proposed in the study by Sastry et al. [22], where modified variants of ARQ were developed to achieve higher throughput under high error rate conditions. The modified

Go-Back-N strategy offers significant performance gains without adding implementation complexity.

In the study by Hara et al. [23], an efficient Stop-and-Wait Automatic Repeat Request (SAW-ARQ) protocol is proposed, which adapts packet length to ensure reliable data transmission at each node. The SAW-ARQ protocol dynamically adjusts the transmission packet length based on varying channel conditions, calculated from the number of NACK (negative acknowledgment) or ACK (acknowledgment) packets received. This proposed protocol effectively reduces the number of retransmissions under both slow and fast Rayleigh fading conditions to achieve high throughput.

In the research conducted by Ghosh et al. [24], the authors analyzed retransmission protocols and proposed a new protocol called Cooperative-Hybrid Automatic Repeat Request for underwater acoustic channels. This approach leverages both the efficiency of incremental redundancy hybrid ARQ, which uses rate-compatible punctured codes, and the reliability of cooperative ARQ.

A study conducted by Kim et al. [25] examines the UCMAC protocol. In this protocol, a resource identifies cooperating nodes, specifies the intended destination nodes, and indicates the distances between them. When errors are received, the receiving node requests retransmissions from cooperating nodes using a nearest-first policy. A designated cooperating node then transmits the data packet that it has saved in its buffer cache, which was successfully received from the original sender or other collaborating nodes. The signaling process and node waiting times are calculated to account for delays caused by cooperation among intermediate nodes.

In the study by Lin et al. [26], an adaptive hybrid ARQ strategy aimed to reduce error rates, the number of retransmissions, and average retransmission times. This strategy employs Q-learning adjustments to determine retransmission frequency and timing, thereby improving transmission reliability. The study also introduces several K-loop transmission schemes, along with delay T and a time overlap interval [T, K], to prevent packet collisions and reduce latency.

Kaythry et al. [27] proposed a recursive Luby transform code-based hybrid automatic repeat request (RLT-HARQ) model to address the challenges of efficient data transmission and reliability. RLT is a form of Luby code characterized by a small degree distribution. The proposed hybrid ARQ scheme combines RLT coding with selective retransmission ARQ, using non-proportional coding. RLT codes are chosen for their minimal complexity in both encoding and decoding processes. The performance of the RLT-HARQ scheme is evaluated considering specific characteristics of the underwater channel.

A multichannel MAC protocol for underwater acoustic channels was introduced by Wei et al. [28], where sensor nodes serve as transmitters. Their synchronized MAC protocol divides time into three phases to enhance system throughput and energy efficiency.

In a study by Zhang et al., [29] proposed a machine learning-based MAC protocol for duplex communication in single-hop underwater acoustic sensor networks. The aim is to improve latency, throughput, and fair access among nodes.

For boundary detection applications in underwater networks using mobile sensors, Ansa et al. [30] proposed a MAC protocol specifically designed for Autonomous Underwater Vehicles (AUVs).

A MAC protocol with variable time slots was introduced by Dong et al. [31]. However, this protocol requires all nodes to synchronize their working times, which results in high energy consumption and consequently reduces sensor node lifespan.

Another recent study by Guo et al. [32] presented a performance analysis of a MAC protocol for underwater acoustic networks. This model takes into account both the unique characteristics of UANs and the diversity of MAC protocols. However, its performance heavily depends on the design of transmission probability models and packet service time models to estimate energy consumption, throughput, and delay.

Alfouzan et al. [33] proposed an energy-conserving MAC protocol that utilizes a layered architecture along with a distributed clustering strategy. This protocol assigns dedicated time slots to individual sensor nodes to minimize the chance of collisions. It requires scheduled coordination for transmission and reception operations.

In the mentioned studies, the focus has primarily been on MAC protocols such as the contention-based, contention-free, and hybrid types. However, these studies have not concentrated on sleep and wake-up strategies designed to reduce repeated packet collisions during retransmissions. In this study, we propose the ARQ_HMAC scheme, a hybrid MAC protocol that combines asynchronous operation with carrier sensing and synchronous TDMA operation, specifically designed for Underwater Acoustic Sensor Networks (UASNs). The performance of the proposed protocol is evaluated through simulations, focusing on two key metrics: the packet retransmission rate due to collisions at the receiving node, and the overall channel throughput. The simulation outcomes are then benchmarked against those of existing MAC protocols, including the asynchronous contention-free protocol (Pure Aloha), the asynchronous contention-free protocol with carrier sensing (Aloha-CSMA), and synchronous time-division protocols such as Slotted-Aloha, Slotted Aloha with CSMA.

III. ARQ_HMAC MAC PROTOCOL DESCRIPTION

In this research, we focus on configuration of a underwater wireless communication network with fixed nodes. The network consists of a variable number of sensor nodes that can be deployed and replaced to perform collect scientific data over a given area. The sensor nodes gather underwater information and transmit this information to a master node as show in Fig. 1. To achieve this objective, sensor nodes is deployed in a specified zones in seabed.

In the ARQ_HMAC MAC protocol, depending on the network's operational state and timing, nodes can operate in three distinct phases: the initialization phase (Fig. 2), the contention-free phase with CSMA (Fig. 3), and the TDMA phase (Fig. 4). The notations used in this research are summarized in Table 1.

TABLE 1. Summary of notations.

Notations	Definition and unit
n	The number of sensor nodes
n_i	The number of time slots prior to i^{th} time slot of node i
t	A specific moment of the network status
t_c	The moment when a packet collision occurs
t_{start_TDMA}	The moment when the nodes switch to TDMA-based operation mode
t_{end_TDMA}	The moment when the nodes end TDMA-based operation mode
t_{max}	Maximum propagation delay (s)
t_i	The moment when a node received ACK packet for itself in the initial phase.
t_{i0}	The moment when a node sent data packet in the initial phase.
t_d	Transmission time of a data packet
t_{ACK}	Transmission time of a ACK packet
t_{NACK}	Transmission time of a NACK packet
t_{guard}	Guard time
t_{sloti}	Time slot for i^{th} node
t_{Di}	Propagation delay from master node to i^{th} node (s)
$t_{time\ out\ Nodei}$	Duration of carrier sensing
t_{TDMA}	Time of TDMA-based operation mode
t_{ACKi}	The moment when a node received
	NACK/xNACK packet.
d_{max}	Maximum communication range (m)
d_i	Distance communication range between master node and the ith node (m)
xACK	ACK packet intended for another station
xNACK	NACK packet intended for another station
xData	Data packet intended for another station

A. INITIAL PHASE (FIG.2)

At the initial stage, after the network infrastructure has been configured, the nodes initiate the network startup process to determine their respective distances to the master node. This is achieved by having each node transmit a packet to the master node according to a predefined schedule (Fig. 2).

To ensure the successful network initialization for each node, the time interval between successive packet transmissions from the nodes is determined based on t_{max}, which represents the maximum propagation delay between the master node and the farthest sensor node. The corresponding waiting time for receiving the Acknowledgement (ACK) packet is also set to t_{max}.

The total duration of the initialization process is given by:

$$t_{initial} = n(2t_{max} + t_d + t_{ACK}) \tag{1}$$

The operational synchronization time of the network nodes is reset after the initialization period $t_{initial}$ and is defined as time t=0.

Based on the time of data packet transmission and the time of ACK packet reception, along with the predefined acoustic propagation speed of 1500 m/s in the underwater environment, each node can determine the acoustic signal propagation delay between itself and the master node in both directions (Fig. 2) as follows:

$$t_{Di} = \frac{t_i - (t_d + t_{ACK}) - t_{i0}}{2} \tag{2}$$

B. CONTENTION-FREE WITH CSMA PHASE

Whenever a node has a packet to transmit, it performs carrier sensing on the channel to determine its current state. Three possible cases may occur:

Case 1: If the channel is free, the node immediately transmits the packet and waits to receive the corresponding ACK/NACK (Fig. 3).

Case 2: If a carrier signal is detected on the channel, the node proceeds to decode the carrier packet. If the decoded carrier is:

- *xACK*, the node will transmit its packet as soon as the channel becomes free.
- *xNACK*, the node will switch to the TDMA-based operating mode.
- *xData*, the node will continue waiting for the corresponding *xACK* or *xNACK* and operate as described above.

Case 3: The node is unable to decode the information from the carrier signal. It will continue to monitor the channel status for a duration $t_{time_out_Nodei}$, during which the following cases may occur:

- *xACK* is received: the node will transmit its packet as soon as the channel becomes free.
- *xNACK* is received: the node will switch to the TDMA-based operating mode.
- *xData* is received: the node will continue waiting for the corresponding *xACK* or *xNACK* and operate as described above.

If the $t_{time_out_Nodei}$ period expires without any carrier detected on the channel, the node will transmit its packet immediately.

After transmitting a data packet, the transmitting node will listen to the channel. If an ACK packet is received, the transmission is considered successful. Conversely, if a Negative Acknowledgement (NACK) packet is received or the waiting time $t_{time_out_Nodei}$ expires without any feedback, the transmission is deemed unsuccessful, and the node must switch to the TDMA-based operating mode.

$$t_{time_out_Nodei} = 2t_{Di} + t_{guard} \tag{3}$$

$$t_{guard} = 2t_d + t_{ACK} \tag{4}$$

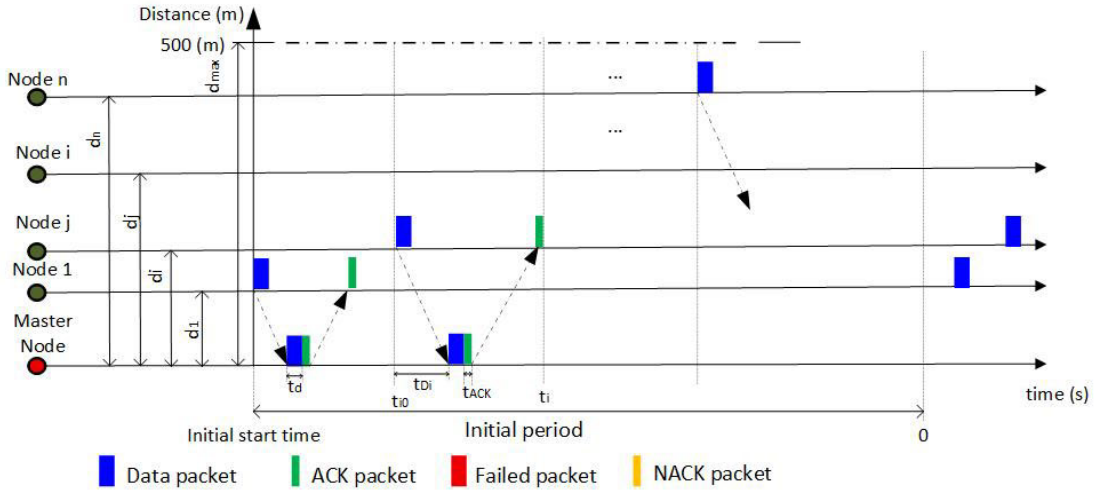


FIGURE 2. Operation of initial phase.

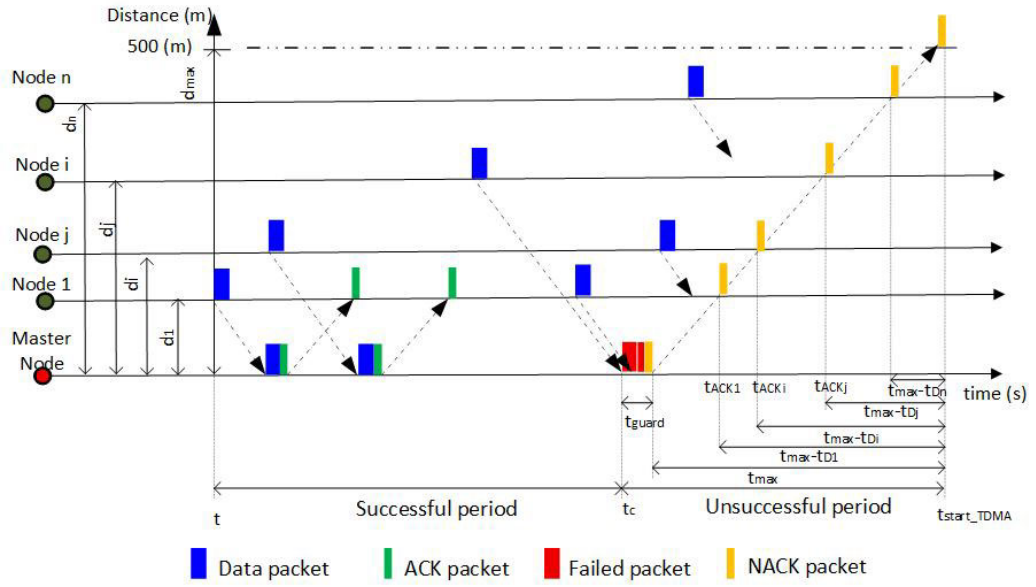


FIGURE 3. A period of Contention-Free with CSMA phase.

1) TDMA-BASED PHASE

In this phase, the nodes in the network operate using the Time Division Multiple Access (TDMA) scheme (Fig. 4), meaning that each node calculates the starting time of TDMA-synchronized operation, denoted as t_{start_TDMA}, and determines its allocated time slot to initiate data transmission precisely at the beginning of that slot.

$$t_{start_TDMA} = t_{ACKi} + t_{max} - t_{Di} \tag{5}$$

In this protocol, when the nodes operate under time slot allocation, the entire channel resource is dedicated to the transmission between a single transmitter–receiver pair. The time slot is designed to be sufficiently long to allow a packet to propagate over the maximum possible transmission distance. Consequently, data transmission in this phase is always successful for the nodes, and therefore, an ACK packet is not required. The time slot duration allocated to each node is denoted as t_{slot}.

To account for potential clock drift in the system, a guard time can be incorporated to slightly extend the slot duration. Moreover, due to the "paradox of residual life" [34], the time slot width t_{slot} is calculated as follows:

$$t_{slot} = t_{max} + t_{guard} \tag{6}$$

The start time of the ith time slot allocated to the ith node, denoted as t_{sloti}, is calculated as follows:

$$t_{sloti} = \begin{cases} t_{start_TDMA} + n_i t_{slot}, & if \quad t_{Di} \; mod \quad t_{slot} = 0 \\ t_{start_TDMA} + n_i t_{slot} + 1, & if \quad t_{Di} \; mod \quad t_{slot} \neq 0 \end{cases} \tag{7}$$

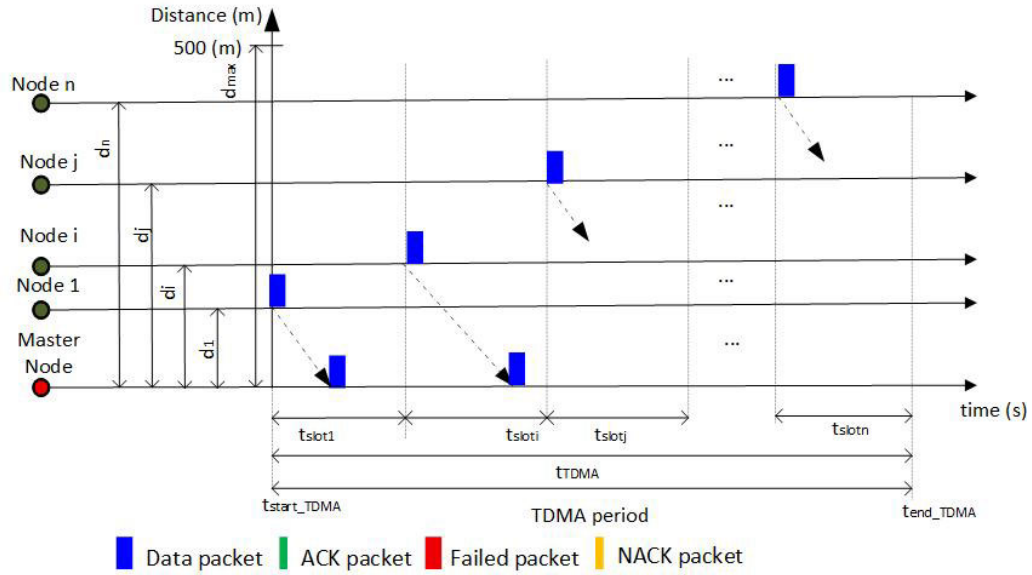


FIGURE 4. A period of TDMA phase.

$$n_i = \frac{t_{Di}}{t_{slot}} \tag{8}$$

The duration during which the network operates based on TDMA slot allocation, denoted as t_{TDMA}, is calculated as follows:e

$$t_{TDMA} = nt_{slot} \tag{9}$$

The network terminates the TDMA operation phase and switch to contention-free operation with CSMA at time t_{end_TDMA}.

$$t_{end_TDMA} = t_{start_TDMA} + t_{TDMA} \tag{10}$$

IV. NETWORK THROUGHPUT ANALYSIS

To analyze the throughput S of the channel in the ARQ_HMAC protocol, G denotes the arrival rate of new packets (offered traffic rate), and t_d as the length of a packet, which has a value of 1. Let t be the arrival time of a packet that senses the channel as idle, given that no other packet is being transmitted

The operations of ARQ_HMAC protocol of a node in time is illustrated in Fig.5.

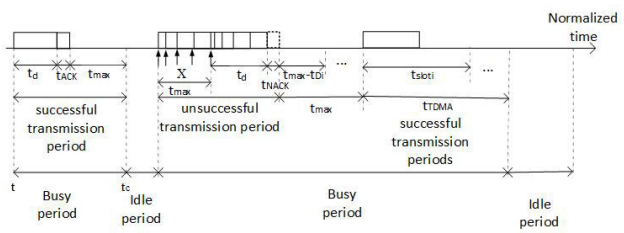


FIGURE 5. A transmission operation of a node.

For the period of contention-free with CSMA, the throughput is denoted S_1. Let $\overline{U}_1$ be the time during a busy period that the channel is used for transmitting data successfully; $\overline{B}_1$ be the average duration of a busy interval; $\overline{I}$ be the average duration of an idle period.

The network throughput for this period is calculated as follows:

$$S_1 = \frac{\overline{U}_1}{\overline{B}_1 + \overline{I}} e^{-2t_{max}G} \tag{11}$$

$$\overline{U}_1 = t_d \tag{12}$$

$$\overline{B}_1 = t_d + t_{ACK} + t_{max} \tag{13}$$

$$\overline{I} = \frac{1}{G} \tag{14}$$

Applying the expressions (12)-(14) for $\overline{U}_1, \overline{B}_1, \overline{I}$ we get S_1 in (11) as follows:

$$S_1 = \frac{\overline{U}_1}{\overline{B}_1 + \overline{I}} e^{-2t_{max}G} = \frac{t_d}{t_d + t_{ACK} + t_{max} + \frac{1}{G}} e^{-2t_{max}G} \tag{15}$$

For the period of TDMA-based phase, the throughput is denoted S_2. Let $\overline{U}_2$ be the number of packets generated during the first t_{max}seconds; all these packets will be successfully transmitted within the t_{TDMA} interval [35].

Let t_c+X be the time of occurrence of the last packet arriving within the interval [t_c, t_c+t_{max}]. The distribution function for X is:

$$\begin{aligned} &F_X(x) Pr X \leq \\ &x = Pr\{no\ arrival\ occurs\ in\ an\ interval\ of\ length\ \ t_{max} - x\} \\ &= exp\{-G(t_{max} - x\}, (x \leq t_{max}) \end{aligned} \tag{16}$$

The average of X is therefore given by

$$\overline{X} = t_{max} - \frac{1}{G}(1 - e^{-t_{max}G}) \tag{17}$$

Let $\overline{B}_2$ denote the average duration of a busy interval, and let $\overline{X}$ represent the expected value of X. The transmission of all packets arriving within the interval $[t_c,\ t_c + X]$ will be completed at time t_c+X+1. After this point, the channel will be perceived as free. All nodes with data packets to transmit within the interval $[t_c + t_{max},\ t$+X+1+$t_{max}]$ will perceive the channel as busy.

$$S_2 = \frac{\overline{U}_2}{\overline{B}_2 + \overline{I}}(1 - e^{-2t_{max}G}) \tag{18}$$

$$\overline{U}_2 = t_d \tag{19}$$

$$\overline{B}_2 = \overline{X} + t_d + t_{NACK} + 2t_{max} + t_{guard} \tag{20}$$

Applying the expression for $\overline{U}_2, \overline{B}_2, \overline{I}$ we get S_2 as follows:

$$S_2 = \frac{t_d}{\overline{X} + t_d + t_{NACK} + 2t_{max} + t_{guard} + \frac{1}{G}}(1 - e^{-2t_{max}G}) \tag{21}$$

Accordingly, the system throughput S is determined by the following formula:

$$S = S_1 + S_2 = \frac{\overline{U}_1}{\overline{B}_1 + \overline{I}}e^{-2t_{max}G} + \frac{\overline{U}_2}{\overline{B}_2 + \overline{I}}(1 - e^{-2t_{max}G}) \tag{22}$$

$$S = \frac{t_d e^{-2t_{max}G}}{t_d + t_{ACK} + t_{max} + \frac{1}{G}} + \frac{t_d(1 - e^{-2t_{max}G})}{t_d + t_{NACK} + 3t_{max} + t_{guard} + \frac{1}{G}e^{-t_{max}G}} \tag{23}$$

V. SIMULATION RESULTS

We conduct simulations to evaluate network performance, comparing the proposed protocol with asynchronous protocols including Pure Aloha, Aloha-CSMA, Slotted Aloha, Slotted Aloha-CSMA, and our proposed ARQ_HMAC MAC protocol. The simulation models the behavior of each node, as illustrated in Fig.1.

The network consists of one master node and ten client nodes, randomly placed at fixed positions on the seabed within an area of 1 km^2. The simulation parameters is showed in Table 2.

TABLE 2. Simulation parameters.

Parameters	Value
Data packet size	1200 bits
Control packet size (ACK/NACK)	120 bits
Maximum transmission range	400 meters
Duration of each simulation	1800 seconds
Time slot duration	0.5 seconds
Acoustic dissemination speed	1500m/s
Bit rate	1200bps

Data packets are generated at random by each client node. The packet generation process follows a uniform distribution over a time interval equal to the simulation duration divided by the number of client nodes.

In our experiments, the offered load per node varies from 20 packets upwards, until the network protocols reach saturation. For each offered load value, we run 10 independent simulations and compute the average performance metrics across the runs, yielding mean values per node. Simulation results are collected to evaluate the performance of each protocol based on the following parameters: The number of successfully transmitted packets per node; The number of packet retransmissions due to collisions per node; Packet delivery ratio.

Fig.6 illustrates the relationship between normalized throughput (S), offered traffic (G), and the maximum transmission range for the proposed ARQ_HMAC MAC protocol. Across all transmission ranges, throughput initially increases with G, reaches a peak, and subsequently stabilizes as G continues to grow. The peak throughput is observed at moderate traffic loads, beyond which the protocol maintains a high and stable throughput without significant degradation, demonstrating its robustness under heavy traffic conditions.

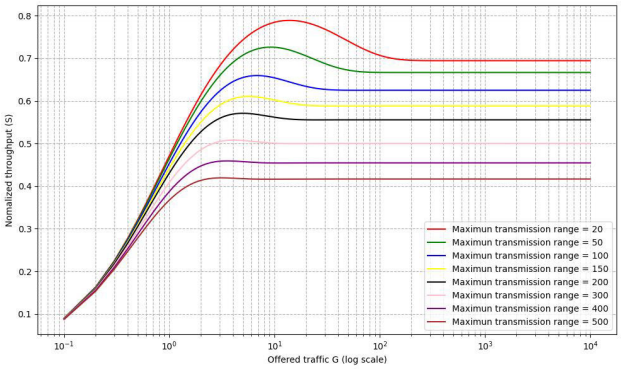


FIGURE 6. The throughput versus the variable of offered traffic and maximum transmission ranges.

It can be observed that shorter maximum transmission ranges yield higher peak throughput values. For example, at a maximum transmission range of 20 m, the throughput approaches 0.78, whereas at 400 m, the maximum throughput is around 0.42. This performance difference is primarily attributed to the increased likelihood of collisions and channel contention when the transmission range is large, as more nodes fall within each other's carrier-sensing and interference zones.

The superior and stable performance of ARQ_HMAC across varying network loads and ranges is the result of its hybrid design. By integrating a contention-free mechanism, the protocol effectively exploits idle channel resources during low traffic periods, maximizing channel utilization. Concurrently, the CSMA-based time-slot allocation significantly reduces collisions when traffic load is high. Additionally, the protocol employs a synchronization mechanism that

accurately calculates and coordinates transmission timings among nodes, ensuring synchronized channel access and minimizing the probability of packet collisions. This combination of idle-channel exploitation, contention reduction, and synchronization enables ARQ_HMAC to maintain higher and more stable throughput compared to conventional MAC protocols, even in scenarios with extended transmission ranges and high offered traffic.

Fig.7 illustrates the variation of normalized throughput (S) of the ARQ_HMAC protocol with respect to different maximum transmission ranges (d_{max}) under various offered loads (G). The results show that as d_{max} increases, the throughput decreases significantly for all values of G. This degradation is primarily attributed to the enlarged communication coverage, which increases the number of nodes within mutual transmission range. Consequently, the probability of packet collisions and the impact of propagation delay both rise, leading to reduced channel utilization efficiency.

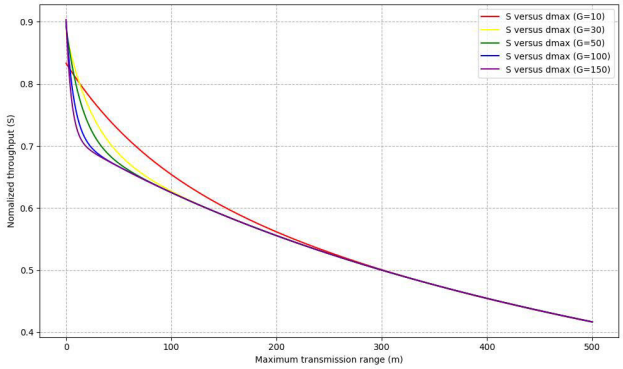


FIGURE 7. The throughput versus diffidence of transmission ranges.

Nevertheless, owing to its hybrid access mechanism combining contention-free MAC operation for efficient channel usage under low contention with CSMA to mitigate collisions under high contention ARQ_HMAC is able to maintain relatively high throughput, especially at small or moderate d_{max} values. Moreover, the throughput for smaller offered loads (e.g., G = 10) remains consistently higher across the entire range of d_{max} compared to larger offered loads, as the lower traffic reduces contention for channel access.

These findings confirm that the performance of ARQ_HMAC is strongly dependent on both the maximum transmission range and the network load. They also highlight the advantage of the proposed hybrid coordination mechanism in mitigating performance degradation across varying transmission distances.

Fig.8 illustrates the number of packet retransmissions for the evaluated MAC protocols as the offered load increases from 20 to 100 packets/node. The ARQ_HMAC protocol demonstrates a consistently lower number of retransmissions compared to the other schemes. This improvement is attributed to its hybrid design, which integrates the contention-free mechanism to exploit idle channel resources and the CSMA mechanism to coordinate time allocation, thereby reducing packet collisions. By effectively combining these two approaches, ARQ_HMAC minimizes retransmission requirements, especially under high traffic loads, highlighting its efficiency in maintaining reliable communication while optimizing channel utilization.

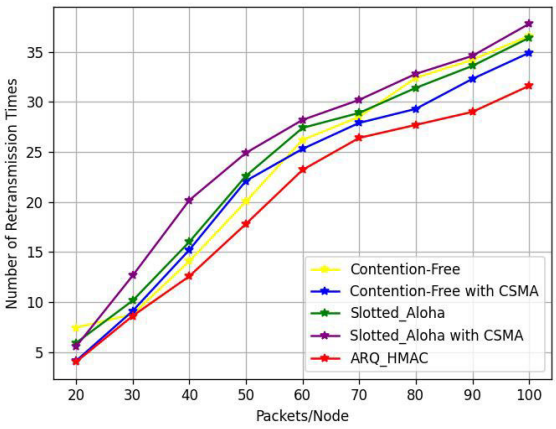


FIGURE 8. Retransmission times.

Fig.9 compares the network throughput of the proposed ARQ_HMAC protocol with four representative MAC schemes: contention-free, contention-free with CSMA, Slotted Aloha, and Slotted Aloha with CSMA. Across the entire range of offered traffic, ARQ_HMAC consistently achieves the highest throughput, demonstrating superior scalability and efficiency. While Slotted Aloha and Slotted Aloha with CSMA experience significant performance degradation due to frequent collisions under moderate to high loads, ARQ_HMAC effectively minimizes packet loss through its integrated ARQ mechanism and optimized hybrid access strategy. The contention-free and contention-free with CSMA protocols exhibit stable performance at low loads but saturate quickly, limiting their throughput under heavy traffic. In contrast, ARQ_HMAC maintains a steady growth in throughput and reaches its peak value at high offered traffic without exhibiting early saturation. This advantage arises from ARQ_HMAC's ability to dynamically balance collision avoidance with flexible time-slot allocation, ensuring efficient channel utilization even in dense network conditions.

Fig.10 illustrates the Packet Delivery Ratio (PDR) performance of the evaluated MAC protocols as the traffic load, expressed in packets per node, increases. Across all traffic conditions, the proposed ARQ_HMAC protocol consistently outperforms the benchmark schemes Contention-Free, Contention-Free with CSMA, Slotted_Aloha, and Slotted_Aloha with CSMA. At a low traffic load (20 packets/node), ARQ_HMAC achieves a PDR of approximately 85%, exceeding the second-best performer, Contention-Free, by nearly 7%. Even at the highest traffic load (100 packets/node), ARQ_HMAC maintains a PDR above 60%, whereas all other schemes degrade to below 55%. This superior performance can be attributed to ARQ_HMAC's

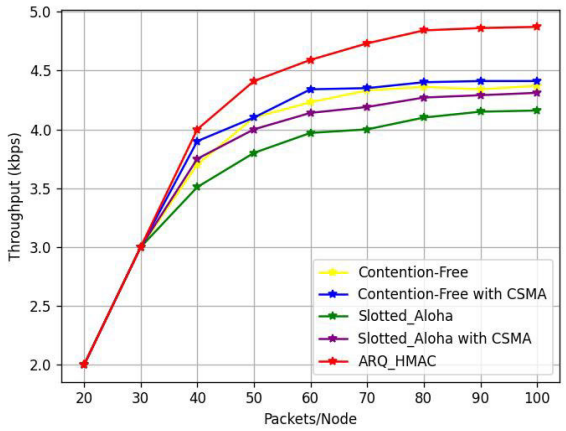


FIGURE 9. Netwwork throughput.

adaptive retransmission mechanism, which enhances delivery reliability under high contention.

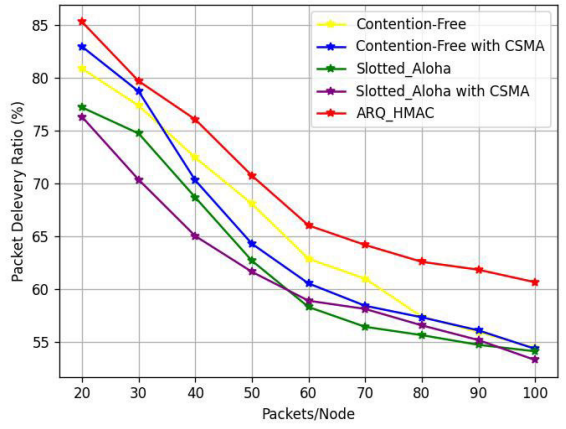


FIGURE 10. Packet delivery ratio.

VI. CONCLUSION

In this study, we proposed the ARQ_HMAC protocol for underwater communication channels. The protocol operates in two modes: an asynchronous contention-free mode and a synchronous TDMA mode. When the network is idle or when no packet collisions are detected, nodes operate asynchronously in a contention-free manner. Upon detecting collisions, nodes switch to synchronous operation. The transition timing to the synchronous state is carefully calculated to prevent packet collisions at the receiver during subsequent retransmissions, thereby reducing the number of retransmissions, improving transmission efficiency, and channel throughput more effectively. Simulation results demonstrate that ARQ_HMAC outperforms both synchronous and asynchronous MAC protocols, including asynchronous contention-free protocols (Pure Aloha, Aloha with CSMA) and synchronous time-division protocols (Slotted Aloha, Slotted Aloha with CSMA) schemes, in terms of network throughput, as shown in Figs. 6, 7, and 9. Furthermore, the protocol significantly reduces packet collisions, leading to fewer retransmissions (Fig. 8) and improving the packet delivery ratio (Fig. 10).

This work can be extended in several research directions, including the consideration of multi-hop networks, energy efficiency, the movement of node with difference depth of sea water, which will be included in our future works.

REFERENCES

[1] I. F. Akyildiz, D. Pompili, and T. Melodia, "Challenges for efficient communication in underwater acoustic sensor networks," *ACM SIGBED Rev.*, vol. 1, no. 2, pp. 3–8, Jul. 2004.

[2] M. Stojanovic and J. Preisig, "Underwater acoustic communication channels: Propagation models and statistical characterization," *IEEE Commun. Mag.*, vol. 47, no. 1, pp. 84–89, Jan. 2009.

[3] M. K. Park and V. Rodoplu, "UWAN-MAC: An energy-efficient MAC protocol for underwater acoustic wireless sensor networks," *IEEE J. Ocean. Eng.*, vol. 32, no. 3, pp. 710–720, Jul. 2007.

[4] N. Chirdchoo, W.-S. Soh, and K. C. Chua, "Aloha-based MAC protocols with collision avoidance for underwater acoustic networks," in *Proc. 26th IEEE Int. Conf. Comput. Commun. (INFOCOM)*, Nov. 2007, pp. 2271–2275.

[5] M. Molins and M. Stojanovic, "Slotted FAMA: A MAC protocol for underwater acoustic networks," in *Proc. Asia–Pacific (OCEANS)*, May 2006, pp. 1–7.

[6] S. Ashraf, M. Gao, Z. Chen, H. Naeem, A. Ahmad, and T. Ahmed, "Underwater pragmatic routing approach through packet reverberation mechanism," *IEEE Access*, vol. 8, pp. 163091–163114, 2020.

[7] K. Chen, M. Ma, E. Cheng, F. Yuan, and W. Su, "A survey on MAC protocols for underwater wireless sensor networks," *IEEE Commun. Surveys Tuts.*, vol. 16, no. 3, pp. 1433–1447, 3rd Quart., 2014.

[8] E. Sozer, M. Stojanovic, and J. G. Proakis, "Underwater acoustic networks," *IEEE J. Ocean. Eng.*, vol. 25, no. 1, pp. 72–83, Jan. 2000.

[9] I. F. Akyildiz, D. Pompili, and T. Melodia, "Underwater acoustic sensor networks: Research challenges," *Ad Hoc Netw.*, vol. 3, no. 3, pp. 257–279, May 2005.

[10] P. Karn, "MACA—A new channel access method for packet radio," in *Proc. 9th ARRL Comput. Netw. Conf.*, 1990, pp. 134–140.

[11] A. Al Guqhaiman, O. Akanbi, A. Aljaedi, and C. E. Chow, "A survey on MAC protocol approaches for underwater wireless sensor networks," *IEEE Sensors J.*, vol. 21, no. 3, pp. 3916–3932, Feb. 2021.

[12] C. L. Fullmer and J. J. Garcia-Luna-Aceves, "Floor acquisition multiple access (FAMA) for packet-radio networks," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 25, no. 4, pp. 262–273, Oct. 1995.

[13] R. Diamant and L. Lampe, "A hybrid spatial reuse MAC protocol for adhoc underwater acoustic communication networks," in *Proc. IEEE Int. Conf. Commun. Workshops*, May 2010, pp. 1–5.

[14] J. Guo, S. Song, J. Liu, H. Chen, J.-H. Cui, and G. Han, "A hybrid NOMA-based MAC protocol for underwater acoustic networks," *IEEE/ACM Trans. Netw.*, vol. 32, no. 2, pp. 1187–1200, Apr. 2024, doi: 10.1109/TNET.2023.3311682.

[15] J.-I. Namgung, S.-Y. Shin, N.-Y. Yun, and S.-H. Park, "Adaptive GTS allocation scheme based on IEEE 802.15.4 for underwater acoustic sensor networks," in *Proc. IEEE/IFIP Int. Conf. Embedded Ubiquitous Comput.*, Dec. 2010, pp. 297–301.

[16] M. K. Watfa, S. Selman, and H. Denkilkian, "UW-MAC: An underwater sensor network MAC protocol," *Int. J. Commun. Syst.*, vol. 23, no. 4, pp. 485–506, Apr. 2010.

[17] A. Roy and N. Sarma, "A synchronous duty-cycled reservation based MAC protocol for underwater wireless sensor networks," *Digit. Commun. Netw.*, vol. 7, no. 3, pp. 385–398, Aug. 2021.

[18] A. Roy and N. Sarma, "Multichannel ordered contention MAC protocol for underwater wireless sensor networks," *Comput. J.*, vol. 64, no. 1, pp. 185–194, Nov. 2019.

[19] A. Roy and N. Sarma, "RPCP-MAC: Receiver preambling with channel polling MAC protocol for underwater wireless sensor networks," *Int. J. Commun. Syst.*, vol. 33, no. 9, p. 4383, Jun. 2020.

[20] H. Zhao, H. Chen, and L. Xie, "A scheduling-based hybrid medium access control protocol for underwater acoustic sensor networks," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Hangzhou, China, Nov. 2023, pp. 639–644.

[21] M. Stojanovic, "Optimization of a data link protocol for an underwater acoustic channel," in *Proc. Eur. Oceans*, 2005, pp. 68–73.
[22] A. Sastry, M. Moeneclaey, J. M. Morris, P. F. Turney, and J. A. Martins, "Improving automatic repeat-request (ARQ) performance on satellite channels under high error rate conditions," *IEEE Trans. Commun.*, vol. COM-23, no. 4, pp. 436–439, Apr. 1975.
[23] S. Hara, A. Ogino, M. Araki, M. Okada, and N. Morinaga, "Throughput performance of SAW-ARQ protocol with adaptive packet length in mobile packet data transmission," *IEEE Trans. Veh. Technol.*, vol. VT-45, no. 3, pp. 561–569, Mar. 1996.
[24] A. Ghosh, J.-W. Lee, and H.-S. Cho, "Throughput and energy efficiency of a cooperative hybrid ARQ protocol for underwater acoustic sensor networks," *Sensors*, vol. 13, no. 11, pp. 15385–15408, Nov. 2013.
[25] H.-W. Kim, T. H. Im, and H.-S. Cho, "UCMAC: A cooperative MAC protocol for underwater wireless sensor networks," *Sensors*, vol. 18, no. 6, p. 1969, Jun. 2018.
[26] S.-Y. Lin, M.-H. Yang, and S. Jia, "An adaptive hybrid automatic repeat request (A-HARQ) scheme based on reinforcement learning," *Electronics*, vol. 12, no. 19, p. 4127, Oct. 2023.
[27] P. Kaythry, R. Kishore, and V. N. Priyanka, "Performance analysis of LT code-based HARQ error control in underwater acoustic sensor networks," *J. Mar. Eng. Technol.*, vol. 21, no. 1, pp. 54–64, Jan. 2022.
[28] Y. Wei and D. Wang, "Graph coloring-based multichannel MAC protocol in distributed underwater acoustic sensor networks," *Frontiers Mar. Sci.*, vol. 9, Jan. 2023, Art. no. 1005959, doi: 10.3389/fmars.2022.1005959.
[29] W. Zhang, J. Li, and Y. Wan, "Machine learning-based performance-efficient MAC protocol for single hop underwater acoustic sensor networks," *J. Grid Comput.*, vol. 20, no. 4, p. 41, 2022, doi: 10.1007/s10723-022-09636-9.
[30] S. S. Ansa, M. Bhavya, and C. Sarang, "Load-adaptive MAC protocol for frontier detection in underwater mobile sensor network," *Ad Hoc Netw.*, vol. 165, Dec. 2024, Art. no. 103641, doi: 10.1016/j.adhoc.2024.103641.
[31] W. Dong, Q. Yang, X. Huang, Y. Chen, S. Sun, and D. Li, "A MAC protocol for underwater sensor networks using parallel transmission variable time slots," *J. Coastal Res.*, vol. 38, no. 2, pp. 449–457, Oct. 2022.
[32] J. Guo, S. Song, J. Liu, H. Chen, Y. Xu, and J.-H. Cui, "Exploring applicable scenarios and boundary of MAC protocols: A MAC performance analysis framework for underwater acoustic networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 12717–12730, Dec. 2024, doi: 10.1109/TMC.2024.3417895.
[33] F. A. Alfouzan, A. Shahrabi, S. M. Ghoreyshi, and T. Boutaleb, "An energy-conserving collision-free MAC protocol for underwater sensor networks," *IEEE Access*, vol. 7, pp. 27155–27171, 2019, doi: 10.1109/ACCESS.2019.2901646.
[34] L. Kleinrock, *Queueing Systems, Volume 1: Theory; Volume 2: Computer Applications*. New York, NY, USA: Wiley Interscience, 1975.
[35] L. Kleinrock and F. Tobagi, "Packet switching in radio channels: Part I-carrier sense multiple-access modes and their throughput-delay characteristics," *IEEE Trans. Commun.*, vol. COM-23, no. 12, pp. 1400–1416, Dec. 1975.



VAN-PHUONG DAO received the master's degree in electronic engineering from Ho Chi Minh City University of Technology and Education, Vietnam, in 2010, and the Ph.D. degree in electronic of telecommunications from Southeast University, China, in 2014. He is currently a Lecturer with Ho Chi Minh City University of Technology and Education. His research interests include MAC protocols of underwater acoustic communication networks, acoustic signal processing, and saving energy for sensor nodes in underwater acoustic communication networks.



THU-HA TRAN received the Ph.D. degree in industrial electronics, automation and telecommunications from Kiev Polytechnic University, Ukraine, in 1996. Since 1997, she has been with Ho Chi Minh City University of Technology and Education, as a Lecturer with the Electrical and Electronics Department. In 2011, she got a title of Associate Professor Ph.D. She spent the time for researching and teaching in the fields, in electronics industrial for using neural networks, fuzzy logic in auto-control, sliding control, PID, robot, voice control, the IoT. In the area of telecommunications, she is conducting research some problems on telex, image processing, noise cancellation in signal transmission systems; and research on signal processing, watermarking video security, and video processing.